



Amrita School of Engineering, Chennai

Course Delivery Plan

Course Code: Course Name	23MEE208: Materials Science	Program	Robotics and Artificial Intelligence
L - T - P - C	2-0-3-3	Semester / Year	III/II
Name of the Faculty	Dr. Sunil Kumar	Pre-requisite	Basic knowledge of chemistry
Course Mentor	Dr. Sunil Kumar	Academic Year	2026 – 27
Course Overview	This course provides an introduction to the structure, properties, and behavior of engineering materials, including metals, polymers, ceramics, composites, and smart materials. It covers crystallography, mechanical properties, strengthening mechanisms, and failure modes such as fracture, fatigue, and creep. Students will also gain exposure to heat treatment processes, material selection using Ashby's chart, and testing methods. Lab sessions offer hands-on experience in mechanical testing and microstructure analysis.		

Course Objectives		Course Outcomes	
1	To impart knowledge on fundamental concepts of metallic materials behaviour and crystal structures.	CO1	Analyze the structure and mechanical properties of engineering materials and apply them to engineering problems.
2	To inculcate the theory of fracture mechanics, fatigue and creep properties.	CO2	Assess different types of ferrous and non-ferrous materials and determine the appropriate material for the given Application.
3	To facilitate an understanding of various strengthening mechanisms, heat treatment, for specific materials and requirements.	CO3	Investigate the effects of heat treatment on microstructures and mechanical properties.
4	To familiarize with the properties and applications of non ferrous materials, composite materials, and advanced materials.	CO4	Understand the characteristics and applications of advanced materials and material selection concepts for Robotics.

Syllabus

Unit 1

Classification of Engineering Materials, Structure of Crystalline Solids - Crystal Systems - unit cells - Metallic Crystal Structures - Miller indices - Crystallographic planes and directions - Linear and Planar Atomic Densities.

Imperfections in Solids: Point – Linear - Interfacial defects - Surface and Volumetric defects.

Mechanical properties: Elastic, Anelastic and Plastic behaviour - Stress-strain curves for Ductile and Brittle alloys – Ductility – Resilience – Toughness - Hardness and testing.

Unit 2

Dislocations and Plastic deformation - Slip phenomenon - Slip in single crystals.

Strengthening mechanisms: grain boundary hardening - solid solution hardening - Hume-Rothery rule - work hardening - Precipitation hardening – recovery recrystallization and grain growth.

Failure of Materials: Ductile and Brittle Fracture - fracture mechanics - Impact fracture - Ductile to brittle transition – Fatigue - Creep properties.

Unit 3

Overview of Heat treatment of steels - Classification of cast iron and steels - properties, microstructures, and applications.

Classification of Non-ferrous alloys properties, microstructures, and applications. Introduction to polymers.

Introduction to composite materials – ceramics - electronic materials - smart materials.

Introduction to material selection process – Ashby's chart. Materials for soft robotics.

Lab Components

Mechanical Testing of materials - Stress-strain curves for Ductile and Brittle alloys - Ductility – Resilience – Toughness - Hardness and testing.

Failure of Materials: Ductile and Brittle Fracture - fracture mechanics - Impact fracture - Fatigue - Creep properties. Heat treatment of steels. Study of microstructure of ferrous and non-ferrous materials.

Text Books

T1. Callister W. D. “Materials Science and Engineering”, 10/e, John Wiley & Sons, 2018.

T2. Avner S. H., “Physical Metallurgy”, 2/e, McGraw Hill Education, 2017.

References

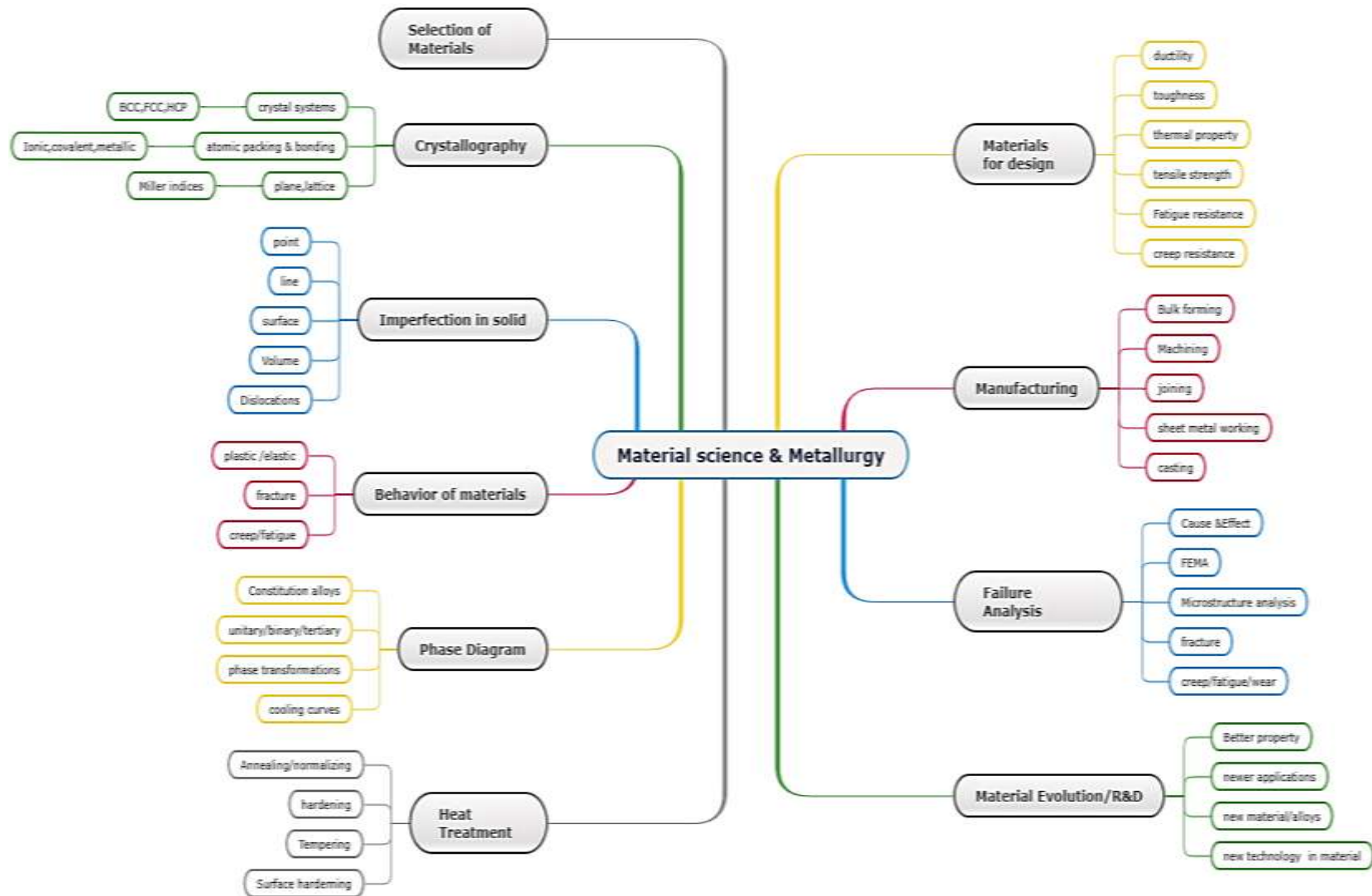
R1. Shackelford J. F., “Introduction to Materials Science for Engineers”, 8/e, Prentice Hall, 2014.

R2. JavedHashemi, Smith F. W., “Foundations of Materials Science and Engineering”, 6/e, McGraw Hill Education, 2022.

R3. Dieter G. E., “Mechanical Metallurgy”, 3/e, Tata McGraw Hill, 2013.

R4. Michael F. Ashby, “Materials Selection in Mechanical Design”, 5/e, Butterworth-Heinemann, 2016.

Concept Map



Evaluation and Grading					
Internal (60)				External (40)	Total
Components		Marks	Total Marks	End Semester = 40	Internal + External = 100
MID Term Exam		30	30		
Continuous Assessments	Quiz (2)	10	30		
	Assignment (1)	5			
	Lab Assessment	10			
	Active Participation in Class / Attendance	5			
Programme Outcome (PO)					
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.				
PO2	Problem analysis: Identity, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.				
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.				
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.				
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.				
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.				
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.				
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.				
PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.				

PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
PSO1	Design and develop cost-effective robotic systems catering to Industrial and Societal requirements.
PSO2	Develop cost-effective, safe, and efficient AI-based automation systems for manufacturing applications, focusing on product development and process improvement.
PSO3	Apply the acquired knowledge and skills in AI to address real-life multidisciplinary engineering problems.

CO – PO Affinity Map

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2				1	1		1	1		2			
CO2	3	3				1	1		1	1		2			
CO3	3	3	1	1		1	1		1	1		2			
CO4	3	3	2	1		1	1		1	1		2			

3 – Strong, 2 Moderate, 1 -weak

Threshold (%)	Target (%)	Percentage of Students		
		Level 1	Level 2	Level 3
50%	55%	50%	60%	70%

Lecture Plan

Class	Topics to be covered	Mode of Teaching	In-Class Activities	Out- Class Activities (E – Ref.)	CO Mapping	Reference
Unit 1						
1 (L)	Classification of Engineering Materials	Smart Board (Flipped Classroom)	Introduction to the Metallurgy and Materials Science	https://www.youtube.com/watch?v=qAeaHYSX0hs https://www.youtube.com/watch?v=KilQCJNucUw https://nptel.ac.in/courses/113/105/113105024/	CO1	T1, T2, R1-R4
2 (L)	Structure of Crystalline Solids - Crystal Systems - unit cells	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=4cRRA8IzU_w https://nptel.ac.in/courses/113/105/113105024/	CO1	T1, T2, R1-R4
3, 4 (L)	Metallic Crystal Structures - Miller indices Crystallographic planes and directions - Linear and Planar Atomic Densities	Smart Board (Flipped Classroom)	Discussion about the basics concept and Solve problems	https://nptel.ac.in/courses/113/105/113105024/ https://www.youtube.com/watch?v=mc-VocwArHY https://www.youtube.com/watch?v=1vsNxxdlcXw	CO1	T1, T2, R1-R4
5 (L)	Imperfections in Solids: Point – Linear - Interfacial defects - Surface and Volumetric defects	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://nptel.ac.in/courses/113/105/113105024/ https://www.youtube.com/watch?v=Z0Ks3wjFnrk	CO1	T1, T2, R1-R4

6 (L)	Mechanical properties: Elastic, Anelastic and Plastic behaviour	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=WxpennU20Ls https://www.youtube.com/watch?v=GyIf_E8vxJw https://www.youtube.com/watch?v=YKpvYF0hVDE	CO1	T1, T2, R1-R4
7, 8 (L)	Stress-strain curves for Ductile and Brittle alloys	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=GyIfE8vxJw https://www.youtube.com/watch?v=Gtd-eLxtP7Y https://www.youtube.com/watch?v=zc3b6LdDFtY	CO1	T1, T2, R1-R4
9, 10 (L)	Ductility – Resilience – Toughness - Hardness and testing	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=OBXXGDyfiN0 https://www.youtube.com/watch?v=TX4umgBk8sQ	CO1	T1, T2, R1-R4
Unit 2						
11, 12 (L)	Dislocations and Plastic deformation - Slip phenomenon - Slip in single crystals	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=laONZJ7YJS4 https://www.youtube.com/watch?v=cpvTwYAUeA8 https://www.youtube.com/watch?v=7qgBMKS_fyo	CO1	T1, T2, R1-R4
13, 14, 15 (L)	Strengthening mechanisms - grain boundary hardening - solid solution hardening - Hume-Rothery rule - work hardening - Precipitation hardening – recovery	Smart Board (Flipped Classroom)	Discussion about the basics concept and Solve problems	https://nptel.ac.in/courses/113/105/113105024/ https://www.youtube.com/watch?v=ROBwsEBrdik https://www.youtube.com/watch?v=j9zwMidxklTg	CO1	T1, T2, R1-R4

	recrystallization and grain growth			https://www.youtube.com/watch?v=14WeQp_UfNo		
16, 17, 18 (L)	Failure of Materials: Ductile and Brittle Fracture - fracture mechanics - Impact fracture - Ductile to brittle transition	Smart Board (Flipped Classroom)	Discussion about the basics concept and Solve problems	https://nptel.ac.in/courses/113/105/113105024/ https://www.youtube.com/watch?v=LRKzOui_kS8 https://www.youtube.com/watch?v=FxQiS8buoDo	CO2	T1, T2, R1-R4
Assignment - 1						
19, 20 (L)	Fatigue - Creep properties	Smart Board (Flipped Classroom)	Discussion about the basics concept and Solve problems	https://www.youtube.com/watch?v=LhUclxBUV_E https://www.youtube.com/watch?v=k8Py4-SdjyU https://www.youtube.com/watch?v=OlexdbPETPw https://www.youtube.com/watch?v=zH05sDLKMoU	CO2	T1, T2, R1-R4
Unit 3						
21 (L)	Overview of Heat treatment of steels	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://nptel.ac.in/courses/113/105/113105024/ https://www.youtube.com/watch?v=ET5igG-Gm30	CO3	T1, T2, R1-R4
22, 23 (L)	Classification of cast iron and steels - properties, microstructures, and applications	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=yGAMQoqzsk8 https://www.youtube.com/watch?v=d4_xSRQxDxs https://www.youtube.com/watch?v=He4bek-XeLc	CO2	T1, T2, R1-R4

				https://www.youtube.com/watch?v=yGAMQoqzsk8		
				https://www.youtube.com/watch?v=FKw231RgFVU		
24, 25, 26 (L)	Classification of Non-ferrous alloys properties, microstructures, and applications	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=NXmv_8FLFsI https://www.youtube.com/watch?v=XN0LF2t-tf0 https://www.youtube.com/watch?v=NXmv_8FLFsI https://www.youtube.com/watch?v=yVvYWLJqVig https://www.youtube.com/watch?v=NXmv_8FLFsI https://www.youtube.com/watch?v=5XM58cK8NBg	CO2	T1, T2, R1-R4

QUIZ - 1

27, 28 (L)	Introduction to polymers. Introduction to composite materials – ceramics - electronic materials - smart materials	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=G41t_0CqPzY https://www.youtube.com/watch?v=X6gga-RCdZI	CO4	T1, T2, R1-R4
29, 30 (L)	Introduction to material selection process – Ashby's chart. Materials for soft robotics	Smart Board (Flipped Classroom)	Discussion about the basics concept	https://www.youtube.com/watch?v=my63D9zG7bc https://www.youtube.com/watch?v=WzN5tPn8toM	CO4	T1, T2, R1-R4

List of Experiments to be Conducted in Materials Science

31-33 (P)	Study of metallurgical microscope and sample preparation	Laboratory	Lab Based Learning	https://youtu.be/vVm0ZW9zBtQ https://youtu.be/fc8zrgYJCJw https://youtu.be/UuHofNW40Yw	CO1	T1, T2, R1-R4
34-39 (P)	Study the microstructure of ferrous materials (low, medium, high carbon steel and high-speed steel, cast irons)	Laboratory	Lab Based Learning	https://www.youtube.com/watch?v=3SPfgT01Vfk https://www.youtube.com/watch?v=sDghCiygFRU	CO2	T1, T2, R1-R4
40-45 (P)	Study the microstructure of Non-ferrous Materials (Aluminium alloys, copper alloys and Al Metal Matrix Composites)	Laboratory	Lab Based Learning	https://www.youtube.com/watch?v=uCiKeUyHsBI https://www.youtube.com/watch?v=are5impXItM	CO2	T1, T2, R1-R4
46-48 (P)	Study of Heat treatment of steels	Laboratory	Lab Based Learning	https://www.youtube.com/watch?v=8H3TiU-Uks8	CO3	T1, T2, R1-R4
49-54 (P)	Study of tensile strength of materials	Laboratory	Lab Based Learning	https://youtu.be/D8U4G5kcpcM	CO3	T1, T2, R1-R4
55-60 (P)	Study of hardness of the materials	Laboratory	Lab Based Learning	https://youtu.be/w8-mAykm00I https://youtu.be/5atPdzh9rK0	CO3	T1, T2, R1-R4
61-66 (P)	Study of fracture mechanics - Impact fracture	Laboratory	Lab Based Learning	https://youtu.be/T3tc33pd3hQ	CO3	T1, T2, R1-R4

67-72 (P)	Study of fatigue strength of materials	Laboratory	Lab Based Learning	https://www.youtube.com/watch?v=ol4i9RTK14s	CO3	T1, T2, R1-R4
73-75 (P)	Study of Creep properties of materials	Laboratory	Lab Based Learning	https://www.youtube.com/watch?v=FztoEU87B90	CO3	T1, T2, R1-R4

Attendance Requirement:

- (i) All students must maintain a minimum of 75% attendance in each course. This requirement is crucial for academic success and eligibility to appear for mid-term and end-semester examinations.
- (ii) Students who fail to meet the 75% attendance threshold will not be permitted to appear for the mid-term and end-semester exams. Requests for exceptions will not be considered under any circumstances.
- (iii) Attendance will be calculated up to three days prior to the commencement of the mid-term and end-semester exams, as per the academic calendar.

Faculty

Course Mentor

Chairperson/ Program Head